ALICE L WALTON SCHOOL OF MEDICINE

Cephalosporins

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Adapted from: Cephalosporins, Vancomycin, and Bacitracin

Listen to this Brick
0:00 / 11:45 1x

Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Describe the mechanism of action of cephalosporins.
2. Categorize the cephalosporin by the generations.
3. Recognize the most common adverse drug reactions associated with cephalosporins.

CASE CONNECTION

DW, an active 65-year-old male, is coughing heavily as he walks slowly toward the exam room. Once inside, he tells you, "This thing has hit me like a train. I can't catch my breath, I'm coughing up some nasty green stuff, and sometimes there's a speck of blood in it. The fevers and chills are exhausting." His respiratory rate is 24/min, pulse 100/min, and pulse ox 92%. He has decreased breath sounds in the right lung base with occasional rhonchi (loud, continous, low pitched, snoring sound).

How will you explain your treatment plan to DW? Consider your answer as you read, and we'll revisit DW at the end of the brick.

GO TO CONCLUSION ↓

What Are Cephalosporins? (LO#1)

The cephalosporins are a class of antibiotic drugs called β-lactams (just like the penicillins). We will discuss what constitutes a β-lactam antibiotic a little later. The cephalosporins are divided into five groups, or generations, based on their effectiveness against different types of bacterial infections. While some are considered narrow-spectrum antibiotics affecting a limited range of organisms, others are considered broad-spectrum antibiotics, covering a wide range of bugs. Cephalosporins can be used to treat many different diseases in the body, from skin infections to pneumonia to urinary tract infections (UTIs).

How Do Cepha

Bacterial Cell Wall

Recall that the bacterial cell wall comprises of peptidoglycans (cross-linked polymer of polysaccharides and peptides). The polysaccharide contains N-acetylglucosamine and N-acetylmuramic acid and an amino-acid peptide is linked to the N-acetylmuramic (NAM) acid sugar terminating in D-alanyl-D-alanine. Penicillin-binding protein (PBP, an enzyme) removes the terminal alanine in forming a cross-link with a nearby peptide providing cell wall rigidity. (Figure 1).

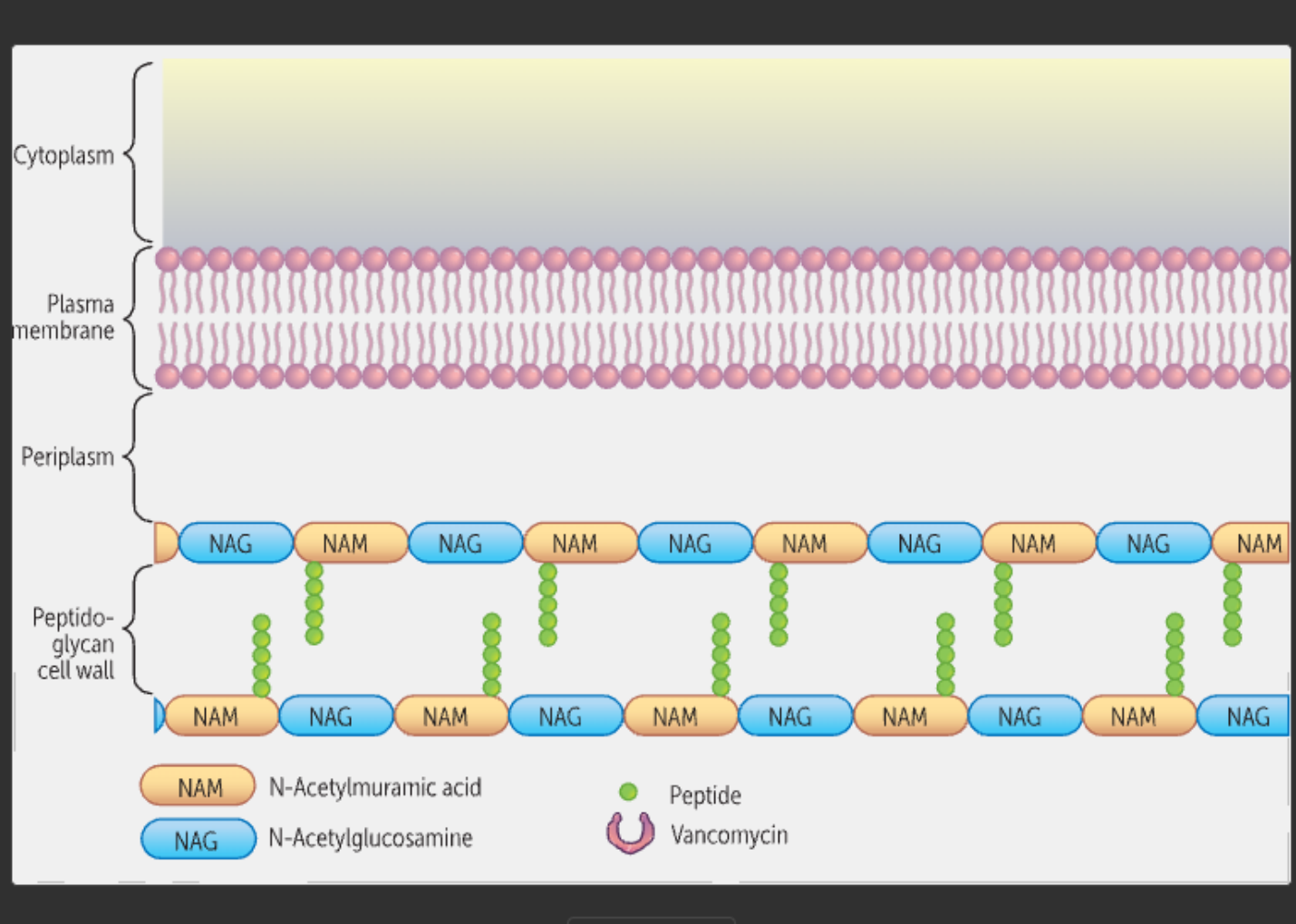


Figure 1

What are the two components of peptidoglycan?

Once these chains of peptidoglycan are constructed, they are cross-linked to each other through peptide chains by enzymes called transpeptidases. This allows multiple layers to develop in the cell wall, forming a rigid network.

Peptidoglycan Cross-Linking Inhibition

Remember that β-lactam antibiotics, like cephalosporins are (Figure 3), structural analogs of the natural D-Ala-D-Ala substrate that covalently bind to the active site of PBPs. Binding of the β-lactam ring inhibits the transpeptidation reaction and halts peptidoglycan synthesis, and the cell dies due to disruption of c… penicillins, note that cephalosporins kill bacterial cells only when they are actively growing and synthesizing cell wall. Thus, most of the cephalosporins are bactericidal antibiotics.

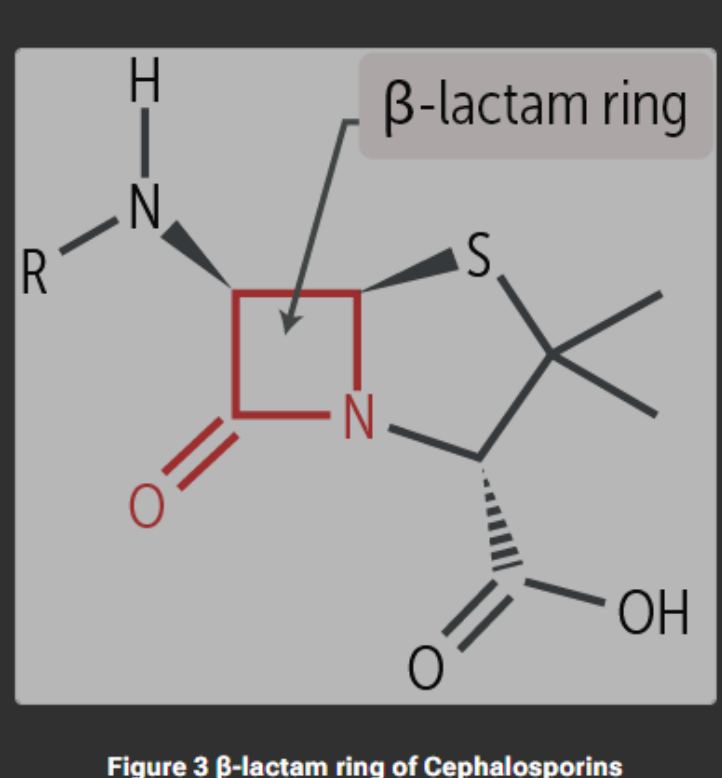


Figure 3 β-lactam ring of Cephalosporins
Credit: ©ScholarRx

What is the primary mechanism of action of cephalosporins?

What are the different Cephalosporin generations? (LO#2)

Cephalosporin generations have been classified based on their antibacterial spectrum of activity, which is covered in later in the course.

Cephalosporins are divided into five generations. The 'generations' represent the timeline for drug development. The activity against specific bacteria varies between the generations, hence the spectrum of activity changes for each generation. But first we are going to break down the generation and list examples of drug commonly used from that generation. The drugs in the various generations are shown in Figure 2: Yes, you should be able to recognize a cephalosporin by its generation!!

First-generation cephalosporins: Cephalexin is a common first-generation cephalosporin. It is used as an oral formulation. **Cefazolin** is another example that is available as an IV formulation (used in surgical prophylaxis).

Mnemonic

What drugs are First-Gen Cephalosporins?

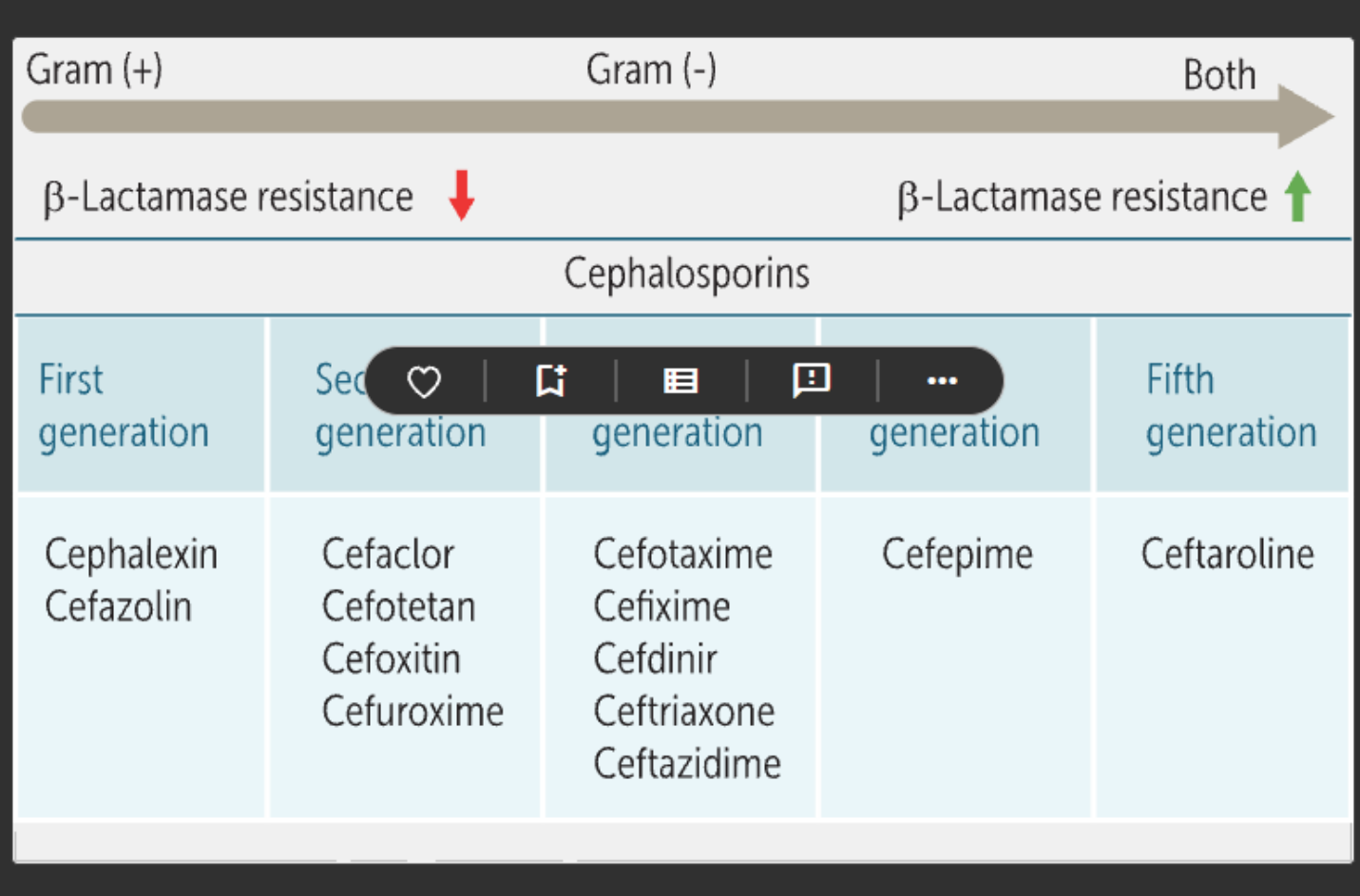
Second-generation cephalosporins: Cefuroxime, Cefoxitin, Cefotetan, and **Cefoxitin** are examples of 2nd generation cephalosporins. They are available as IV and oral dosage forms. *(All of these drugs have either -fo, -fur in the name)*

Third-generation cephalosporins: The most commonly used thrid-generation cephalosporin is **Ceftriaxone** (tri for 3rd generation). The other examples include **Cefotaxime, Cefdinir, Ceftadizime.** *(All the third generation have -ta, -di, -tri, in the name)*

The only **fourth-generation cephalosporin** is **Cefepime**.

Fortunately, **Fifth-generation cephalosporin** only has one drug which is **Ceftaroline.**

The bacterial coverage of these drugs will be discussed in a coverage bricks.


Gram (+)
Gram (-)
Both
β-Lactamase resistance ↓
β-Lactamase resistance ↑
Cephalosporins

First generation	Sec… generation	generation	generation	Fifth generation
Cephalexin Cefazolin	Cefaclor Cefotetan Cefoxitin Cefuroxime	Cefotaxime Cefixime Cefdinir Ceftriaxone Ceftazidime	Cefepime	Ceftaroline

QUIZ

Figure 2
Credit: ©ScholarRx

What Are the Adverse Effects associated with Cephalosporin use? (LO#4)

Although they are responsible for fewer hypersensitivity reactions than penicillins, some cephalosporins can still cause allergic and disulfiram-like reactions and precipitate autoimmune **hemolytic anemia**. They do exhibit some cross-reactivity with penicillins, which means they should not be given to patients with an anaphylactic response to penicillin. Cephalosporins can also damage the kidneys if given together with aminoglycosides, which are notoriously nephrotoxic.

CLINICAL CORRELATION

Cephalosporin-induc… a known possible adverse effect of cephalosporin use, most commonly caused by concomitant ingestion of alcohol. These reactions are called "disulfiram-like" for their resemblance to a reaction caused by the use of disulfiram in the management of chronic alcoholism. A disulfiram reaction is characterized by symptoms such as severe flushing, hypotension, tachycardia, nausea, and vomiting. CIDLR can mimic these symptoms, ranging in severity from mild facial flushing or nausea to life-threatening angioedema or shock. It is of paramount importance to strongly caution patients against the consumption of alcohol while taking cephalosporins, as the combination can have fatal consequences.

Cephalosporins should never be paired with what other class of antibiotics out of concern for nephrotoxicity?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to DW

You explain, "I think you have pneumonia. An x-ray will definitively make the diagnosis, but I'm sure enough that I can already lay out a treatment plan. I'd like to get you started on antibiotics, two different ones, plus oxygen to help your breathing right away. All of this means that we need to get you admitted to the hospital. I know you like to stay active, and you'd probably rather be at home, but I can't readily get you all of these things at home." DW agrees to be admitted. You suspect *Streptococcus pneumoniae* is the cause, although other bacteria are possible. You verify that he does not have allergies, and you write orders for ceftriaxone.

Summary

- Cephalosporins are a class of β-lactam antibiotics, divided into five generations based on their effectiveness against different types of bacterial infections.
- These drugs kill bacteria by altering the cell wall, specifically by inhibiting the transpeptidation reaction, halting peptidoglycan synthesis.
- The spectrum of activity varies across the generations, with first-generation cephalosporins primarily covering gram-positive cocci, and later generations expanding coverage to gram-negative organisms.
- Fifth-generation cephalosporins, such as Ceftaroline, have activity against MRSA.
- Cephalosporins can cause allergic and disulfiram-like reactions, precipitate autoimmune hemolytic anemia, and can damage the kidneys if given together with aminoglycosides.
- Patients with an … hould avoid cephalosporins du…

Review Questions

1. What is the mechanism of action of cephalosporins?

A. ○ They activate autolytic enzymes.
B. ○ They bind to penicillin-binding protein 3.
C. ○ They inhibit the NAM–NAG unit from being added to the peptidoglycan chain.
D. ● They inhibit the transpeptidase that is responsible for the cross-linking of peptidoglycan.
E. ○ They prevent NAM from joining bactoprenol.

Hide Explanation

The correct answer is they inhibit the transpeptidase that is responsible for the cross-linking of peptidoglycan (D). Cephalosporins interfere with cross-linking between the peptidoglycan chains, which promotes rigidity of the cell wall necessary to prevent rupture from intracellular pressure. They activate autolytic enzymes (A) is one of the mechanisms by which penicillins exert their effect. They bind to penicillin-binding protein 3 (B) is the mechanism of action for aztreonam. They inhibit the NAM–NAG unit from being added to the growing peptidoglycan chain (C) is the mechanism of action of vancomycin. They prevent NAM from joining bactoprenol (E) is the mechanism of action of bacitracin.

2. Ceftaroline belongs to which cephalosporin generation?

A. ○ 1st
B. ○ 2nd
C. ○ 3rd
D. ○ 4th
E. ● 5th

Hide Explanation

The correct answer is that Ceftaroline is a fifth-generation cephalosporin.

How would you rate this Brick?

Review Learning Objectives

Check off th… the concept

Hide Explanation

The correct answer is that Ceftaroline is a fifth-generation cephalosporin.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the mechanism of action of cephalosporins.~~
02 ☑ ~~Categorize the cephalosporin by the generations.~~
03 ☑ ~~Recognize the most common adverse drug reactions associated with cephalosporins.~~

Objective Confidence: 3/3